17

Hopping of light atoms

Important technological problems involve the hopping of light atoms within materials comprising relatively heavy atoms. For example, interstitial diffusion of hydrogen in transition metals is a step in these materials' hydrogen embrittlement. Such hopping is similar to electronic small-polaron hopping in that the masses of the diffusing species are smaller than those of the host. The temperature dependence of a light atom's jump rate is even qualitatively similar to that of an electronic small polaron. Nonetheless, the mass of a hydrogen atom is very much larger than that of an electronic carrier. As will become evident this large mass difference requires generalizing the small-polaronic treatment of hopping to render it applicable to light-atom diffusion. As a result the interpretation of the temperature dependence of a light-atom's jump rate generally differs from that for an electronic small polaron. Most markedly, the resulting light-atom jump rate garners a distinguishing dependence on the atom's isotopic mass.

17.1 High-temperature diffusion: excited coincidences and non-Condon transfers

As described in Chapter 11, a small-polaron's hopping is dependent on the motions of the host material's atoms. At high temperatures these motions become classical. Then a light diffusing particle can only hop when the host atoms' movements bring the energies associated with the diffuser occupying initial and final sites into momentary coincidence.

Increasing the mass of the diffusing particle from that of an electronic carrier to that of a light atom greatly reduces the separations between the energy levels of its bound states. In particular, as illustrated in Fig. 17.1, many energy levels can then be supported within the dual potential wells of a symmetric coincidence configuration. In fact these energy separations will generally be comparable to the thermal energy κT at room temperature. Thus light atoms are generally afforded multiple parallel paths via which they can transfer between sites involved in a coincidence.

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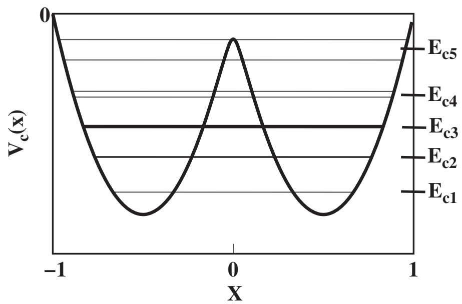

Fig. 17.1 The potential energy for a symmetric two-site coincidence $V_c(x)$ is plotted against a diffusing atom's position x. Even for a very light diffusing atom the separations between successive coincidence energy levels, E_{c1}, E_{c2}, E_{c3}, E_{c4} and E_{c5}, are often small enough for multiple coincidence levels to be thermally accessible at room temperature. The tunnel splitting between each coincidence's two states increases very sharply with coincidence energy. Thus the tunnel splittings in this illustration are only discernable for E_{c4} and E_{c5}.

As described in Sections 11.3–11.6 and depicted in Fig. 11.7 the probability of a diffusing particle availing itself of the transport opportunity offered by the occurrence of a coincidence event depends on the magnitude of the associated transfer energy. For electronic hopping the so-called "Condon approximation", ignoring the dependence of transfer energies on coincidence conditions, is generally invoked. Then the transfer energy for a given electronic carrier is treated as a constant, independent of atomic deformations.

By contrast, the Condon approximation is strongly violated for light-atom diffusion. As illustrated in Fig. 17.1, the transfer energies of light atoms' coincident states, essentially half their tunneling splittings, rise very strongly with coincidence energy. This feature results from a light atom's mass being much larger than that of an electronic carrier. For example, the transfer energy of a particle of mass M tunneling at a coincidence whose energy is E_c below the peak of a barrier of curvature $-k_b$ is $t_{g,g'} \propto \exp[-\pi E_c(M/k_b)^{1/2}/\hbar]$. Thus, the transfer energy's fall with increasing E_c becomes more severe as M is increased: $\partial \ln t_{g,g'}/\partial E_c \sim -\pi(M/k_b)^{1/2}/\hbar$.

Light-atom diffusion is distinguished from electronic hopping by (1) the availability of parallel transfer channels whose (2) transfer energies depend very strongly on coincidence conditions. As a result, unlike the situation for electronic hopping, the predominant hopping of a light atom is not generally associated with the lowest-energy level of the minimum-energy coincidence configuration (Emin *et al.*, 1979). In particular, increasing the mass of the diffuser slows its transfer via low-energy coincidences while increasing the accessibility of higher-energy coincidences with much faster transfer.

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Numerical models have been used to study the hopping of hydrogen and its isotopes between shallow interstitial positions in bcc metals (Emin *et al*., 1979). Transfer energies for low-lying coincidences are found to be so small that transport through them is severely limited. As a result, room-temperature diffusion is dominated by transfer through excited coincidences. The diffusion constant's small activation energy is then just the difference between the energy of the lower of the pair of excited coincidence states and the ground-state energy of an interstitial hydrogen atom. Moreover, excited coincidences' transfer energies are often large enough for the hopping to be adiabatic. In these instances, as described in Chapter 11, an interstitial hydrogen atom hops whenever motions of the transition-metal atoms present it with such excited coincidences. Then the pre-exponential factor of the hydrogen diffusion constant depends on only the motions of the host atoms. In particular, the diffusion constant's pre-exponential factor becomes just $\nu a^2 \approx 10^{-3}$ cm^2/sec, where ν is comparable to the Debye temperature of the host metal and a represents the jump distance.

Diffusion through excited coincidences is progressively frozen out as the temperature is reduced. Concomitantly the diffusion constant falls and its temperature dependence gradually weakens thereby producing non-Arrhenius behavior. At lower temperatures the relatively slow residual semiclassical diffusion is associated with only the diffuser's lowest-energy coincidence state.

Early work ignored the possibilities of excited coincidences and breakdowns of the Condon approximation (Flynn and Stoneham, 1970). Thus the diffusion of interstitial atoms in metals was treated like the hopping of electronic small polarons. As depicted in Fig. 11.2, the hopping of an electronic small polaron also displays non-Arrhenius behavior. However, an electronic small polaron's non-Arrhenius behavior occurs well below the host solid's Debye temperature where multi-acoustic-phonon jump processes are frozen out. At very low temperatures in an ideal crystal this freeze-out process culminates with small-polaron hopping occurring primarily via a two-phonon process in which one phonon is absorbed and another phonon of essentially equal energy is emitted (Holstein, 1959b). With the diffuser interacting with acoustic phonons the low-temperature limit of this two-phonon jump rate is proportional to T^7 (Flynn and Stoneham, 1970). This low-temperature behavior only becomes distinguishable from higher-temperature non-Arrhenius behavior when the temperature is lowered well below the host solid's Debye temperature.

17.2 Isotope dependences of light-atoms' high-temperature diffusion constants

Changing isotope significantly alters a light-atom's mass. Deuterium and tritium have double and triple the mass of simple hydrogen, respectively. As such, altering a

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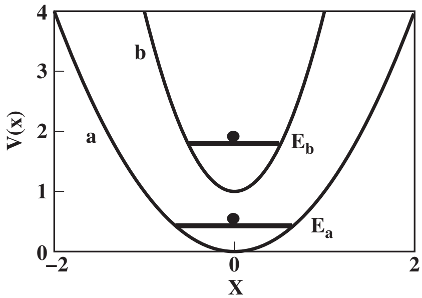

Fig. 17.2 Curves *a* and *b* illustrate the potential energy in the *x*-direction for two configurations of the host material's atoms. Host atoms' movements generally shift both the potential well's minimum and its curvature. The dependence of the diffusing atom's energy, E_a or E_b, on its mass enters through the energy level's dependence on the potential well's curvature.

light-atom's isotope provides a distinctive probe of the light-atom's diffusion process. Indeed, the isotope dependences of light-atoms' diffusion constants can distinguish the mechanism of their hopping from those of electronic small polarons and of classical heavy atoms.

Altering the positions of atoms of the host lattice changes the potential well occupied by the diffusing light atom. As illustrated in Fig. 17.2 modulations of host-atoms' positions generally shift the minimum and the curvature of the potential well containing the diffusing particle. The energy of the light atom changes as its potential well is modified. In particular, Fig. 17.2 shows that the amount that a light atom's energy level is elevated above the potential well's minimum increases as the potential well's curvature increases. The contribution to the total energy shift arising from altering the minimum of the host-atom's potential well is independent of the host-atom's mass. However, the contribution arising from altering the potential well's curvature depends on the diffuser's mass.

Atom–phonon interactions, analogous to electronic polarons' electron–phonon interactions, measure the sensitivities of light-atom's energy levels to shifts of host-atoms' positions. Thus the strengths of these interactions generally fall as a diffusing atom's isotopic mass is increased since heavier isotopes are less sensitive to changes of a potential well's curvature than are lighter isotopes.

The strengths of atom–phonon interactions govern the contribution of host-atom displacements to a light-atom's self-trapping. Since the atom–phonon interaction strength decreases with increasing isotopic mass, the diffusing atom's self-trapping energy also decreases with increasing isotope mass. By itself, this effect would cause heavier isotopes to diffuse more rapidly than lighter isotopes. In particular, the activation energy of an electronic polaron's non-adiabatic semiclassical hopping generally falls as its self-trapping energy falls. In other words, treating a light-atom's hopping as simple non-adiabatic small-polaron hopping implies that the activation energy for the light-atom's hopping will fall as its isotopic mass is increased.

However, as stressed in Section 17.1, the energetic separations between a light-atom's lowest-energy coincidence and its excited coincidences are much less than those for an electronic small polaron. Thus light-atoms' excited coincidences can become thermally accessible even though electronic carriers' excited coincidences are generally ignorable. It is therefore appropriate to consider the diffusion constant for a light atom's thermally activated semiclassical hopping to be a generalization of that for an electronic small polaron:

$$D = va^2 \sum_i P_i \exp\left(-\frac{E_{A,i}}{\kappa T}\right), \tag{17.1}$$

where v is the characteristic vibration frequency of the host atoms and a is the diffusing light-atom's jump distance. Here P_i represents the probability that a diffusing atom hops when presented with the i-th coincidence and $E_{A,i}$ denotes the net energy of that coincidence relative to that of the self-trapped ground-state.

The transfer probability P_i rises strongly with the electronic transfer energy (cf. Fig. 11.7). In addition, as illustrated in Fig. 17.1, a light-atom's excited coincidences have larger transfer energies than those of lower-energy coincidences. Thus relatively large values of P_i tend to be associated with large values of $E_{A,i}$. As a result, when excited coincidences become thermally accessible they tend to dominate a light-atom's high-temperature semiclassical diffusion.

Figure 17.3 illustrates the isotope dependence found from models of high-temperature semiclassical diffusion of interstitial hydrogen in bcc metals (Emin *et al.*, 1979). The highest-temperature diffusion is dominated by coincidences whose transfer energies are large enough to yield adiabatic motion, $P_i \approx 1$. As such, the pre-exponential factor of the diffusion constant is just va^2. The associated activation energy rises as its diffusing isotope's mass increases. This behavior occurs because a larger isotopic mass necessitates forming a higher-energy coincidence to produce a large-enough transfer energy to enable an adiabatic hop.

These features distinguish semiclassical light-atom diffusion from both classical and conventional semiclassical small-polaron hopping. In particular, the diffusion constant's pre-exponential factor being independent of the diffusing light atom's isotopic mass differs from the prediction for classical diffusion. For classical diffusion the diffusion constant's pre-exponential factor is inversely proportional to the square root of the diffuser's mass. In addition, the increase of a diffusion constant's activation energy with its isotope's mass is opposite to the predictions for conventional small-polaron hopping.

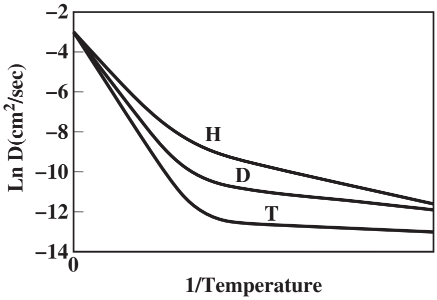

Fig. 17.3 The semiclassical diffusion constants of hydrogen H, deuterium D and tritium T are drawn versus reciprocal temperature in arbitrary units. The three diffusion constants approach their common adiabatic limit $va^2 \approx 10^{-3}$ cm^2/sec at high temperatures. High-temperature diffusion is dominated by adiabatic hopping through excited coincidences. Then the activation energies rise as the isotopic mass is increased. At somewhat lower temperatures diffusion is dominated by non-adiabatic jumps through their lowest-energy coincidences. Then the activation energies fall as the isotopic mass is increased.

At the lowest temperatures of Fig. 17.3 the excited coincidences are frozen out. Then the predominant semiclassical jumps result from their lowest-energy coincidences. As such, a light-atom's semiclassical hop becomes analogous to that of a conventional small polaron. Since the masses of even light atoms are huge relative to those of electronic carriers these hops are generally non-adiabatic. As a result both the diffusion constant's activation energy and its pre-exponential factor decrease with increasing isotopic mass. Figure 17.3 contrasts this low-temperature behavior with the high-temperature behavior where excited coincidences dominate the light atom's diffusion.

17.3 Isotope dependences of light-atoms' low-temperature diffusion constants

Lowering the temperature progressively freezes out excited coincidences and host-atoms' classical motions. At very low temperatures a light-atom's hopping becomes similar to the non-semiclassical hopping of an electronic small polaron. The relatively small masses of electronic carriers compared with those of even light atoms suggests that their transfer energies will generally be small enough to render light-atom's hopping non-adiabatic. Thus a light-atom's low-temperature diffusion becomes formally equivalent to that of a non-adiabatic electronic small-polaron hopping.

As illustrated in panel *a* of Fig. 11.1, low-temperature hopping of an electronic small polaron requires quantum tunneling of both the electronic carrier and the atoms of the host solid. Similarly the low-temperature non-adiabatic small-polaronic

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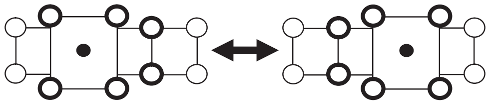

Fig. 17.4 The low-temperature diffusion of a light interstitial atom (solid black dot) occurs as it and surrounding atoms of the host material (open circles) alter their equilibrium positions. For clarity darkened open circles represent those host atoms whose equilibrium positions are shifted in the illustrated jump process.

hopping of a light atom requires tunneling of both the diffusing light atom and the host material's atoms. The initial and final equilibrium locations of the atoms involved in such a tunneling process are illustrated in Fig. 17.4. A light-atom's low-temperature non-adiabatic jump rate is proportional to the absolute squares of both (1) the diffusing atom's transfer energy and (2) the overlap between the wavefunctions for the host atoms' vibrations before and after the hop (Emin, 1975a). Both factors depend on the mass of the diffusing light atom.

Displacements of the equilibrium positions of host atoms surrounding a diffusing light atom depend on its atom–phonon interactions. As discussed in Section 17.2, these interactions measure the sensitivity of the light-atom's energy to changes of the host-atoms' positions. As such, the strengths of these interactions vary inversely with the light-atom's isotopic mass. Thus shifts of the equilibrium positions of host atoms induced by the light-atom's presence decrease upon increasing its isotopic mass. Concomitantly, the depth of the potential well binding the light atom falls as its isotopic mass is increased.

The isotope dependence of a light-atom's low-temperature non-adiabatic small-polaron-like diffusion constant is determined by the isotopic dependences of two factors. The first factor is the absolute square of the overlap between the wavefunctions describing host-atoms' harmonic vibrations about the equilibrium positions they assume in response to a light-atom's hop occupying the hop's initial and final sites. This factor increases as the light-atom's isotopic mass increases because the shifts of host-atoms' equilibrium positions then decrease. The second factor is the absolute square of the light-atom's transfer energy. The depths of the potential wells between which the light atom tunnels decrease as its isotopic mass increases. Nonetheless, the efficacy of the light-atom's tunneling tends to decrease as its isotopic mass increases. All told, the isotope dependence of a light-atom's low-temperature non-adiabatic diffusion is determined by competing effects.

As indicated by the illustration of Fig. 17.3, the isotope dependence of a light-atom's diffusion constant generally changes with temperature. The isotope dependence of a light-atom's diffusion constant may even reverse with falling temperature.

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